# HUY NGUYEN

SYSTEMS SOLUTION ENGINEER

work@huynguyen.email
huynguyen.space
linkedin.com/in/huy-nguyen-au
github.com/HuyNguyenAu

## OVERVIEW

Full stack software engineer building cloud native services, edge systems, and web apps in C#/.NET, React/TypeScript, and Azure, from design through to production. 6+ years of experience, including platforms behind national fleet electrification rollouts for Woolworths, RACV, SG Fleet, and Team Global Express.

## EXPERIENCE

**Full Stack Software Engineer**

JET Charge, Melbourne | Oct 2021 – Jun 2026

- **Multi Tenant Billing & Payments:** Built an independent billing engine from scratch in ASP.NET Core with Stripe Connect split payouts, per session metering, and time of use rate pricing. Generated $100k - $300k a year in recurring software revenue, centralised multi tenant authorisation in Entity Framework, and gave fleet managers self service portals in React and TypeScript.
- **Remote Setup & Device Sync (ZTC):** Built a visual pre configuration web tool in React and an automated boot sync service, cutting on site technician time from 6 hours down to 2 - 3.5 hours (zero when pre paired in the warehouse). Decoupled cloud and firmware schemas for independent deployments and added a pointer table cache that cut device connection latency from >30s to <3s and dropped storage costs by 5% - 10%.
- **Edge Energy Management (CORE v2):** Built an offline first C# / .NET daemon on Linux gateways that balances charger loads in real time over Modbus and OCPP, allowing sites to power up to 5x more chargers on existing supplies without grid upgrades. Created an embedded Next.js local dashboard so electricians can commission sites on a tablet without software engineers on site.
- **Delivery & Team Leadership:** Maintained Azure DevOps CI/CD pipelines deploying infrastructure via Bicep to separate per environment Azure subscriptions. Wrote test suites in xUnit, NUnit, and Vitest gated by SonarQube and Roslyn analyzers, handled on call production triage with Application Insights and KQL, and led design sessions, code reviews, and sprint planning for squads of 3 - 5 engineers.

## EDUCATION

**Computer and Network Engineering**

RMIT

Mar 2016 – Mar 2020

## SKILLS

**Backend:** C#, .NET / .NET Core, ASP.NET Core Web API, Entity Framework, REST APIs, Python

**Cloud:** Azure Functions, App Service, Container Apps, Service Bus, Event Grid, Bicep

**Data:** SQL Server / Azure SQL, PostgreSQL, Cosmos DB, Table and Blob Storage

**Frontend:** React, Next.js, TypeScript, Zustand, React Context, Vite, ShadCN UI

**DevOps & Quality:** Azure DevOps Pipelines, xUnit, NUnit, MSTest, Jest, Vitest, SonarQube / SonarCloud,

**Observability:** Application Insights, Azure Monitor, Log Analytics (KQL)

**Integrations:** Stripe Connect, WebSockets, KYC APIs, Modbus RTU / TCP, OCPP 1.6J / 2.0.1

**IT Specialist**
Energy Locals, Melbourne | Oct 2020 – Oct 2021

- **Customer Onboarding Platform:** Built a white label onboarding platform in Laravel and Bootstrap with automated multi tenant database setup, cutting partner onboarding time and processing over 10,000 signups. Integrated Westpac and Equifax KYC APIs, dropping identity checks from 2 - 3 days to under 60 seconds.
- **Wholesale Settlement Reconciliation:** Automated daily settlement auditing in Python and SQL between AEMO market data and customer interval meters to catch variances early and built calculation pipelines for time of use and pass through retail tariffs.
- **Infrastructure & ETL Automation:** Built automated C# and SQL ETL pipelines that saved analysts 10 - 15 hours a week of manual spreadsheet reporting. Wrote Bash and PowerShell scripts to automate developer machine and server setup, reducing onboarding from 2 days to under 2 hours.

**Graduate IT Specialist**
Energy Locals, Melbourne | Sep 2019 – Oct 2020

- **Market Data & Settlement Automation:** Automated daily verification of AEMO wholesale settlement runs using Python and SQL to verify billing calculations and catch discrepancies early and built ingestion feeds for electricity spot prices and regional energy demand data to support daily trading and forecasting.
- **Systems & Infrastructure Automation:** Wrote PowerShell, Bash, and Python scripts for automated database backups and recovery testing and created workstation configuration scripts to streamline new developer onboarding.

## SYSTEMS & APPLIED AI RESEARCH (OPEN SOURCE)

- **Amber 8 (Rust):** An assembly virtual machine that brings the messy real world into the deterministic domain of software.
- **Wayfare (C#):** A coding agent bounded by a finite state machine to make agent workflows predictable, reproducible, and easy to debug.
- **LLM Native Virtual Machine (Rust):** An assembly like virtual machine that breaks prompts into small atomic instructions executed sequentially, helping smaller local models stay accurate without getting lost.
- **Lox Lang AI (C#):** A bytecode interpreter and AST evaluator used to explore AI-generated and AI-optimised execution routines.
- **GPT Decoder (Python):** A minimal transformer inference engine written from scratch in Python and PyTorch to experiment with KV caching and local model execution without cloud dependencies.
- **Wobbly ML (Python):** A small machine learning library built from first principles in C# to understand backpropagation and gradient descent without external dependencies.